

Foto: Dr. Matthias Otte

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EUROfusion

HELMHOLTZ

RESEARCH FOR GRAND CHALLENGES



P. Scholl^{1,2}

IMPURITY TRANSPORT AND RADIATION AT THE STELLARATOR WENDELSTEIN 7-X

Promotionskolloquium

March 21, 2026

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Domstr. 11, D-17489 Greifswald, Germany

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Real-Time Radiation Feedback Control

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MOTIVATION

NUCLEAR FUSION: ENERGY CHALLENGE

Lawson criterion for net energy gain:

$$n_e T \tau_E \geq \frac{12 f_{\text{tot}}}{\langle \sigma_{\text{DT}} v \rangle f_H^2 E_\alpha - 4 L_Z(T)} T^2$$

$$\gtrsim 3 \times 10^{21} \text{keVs/m}^3$$

(requires temperatures $T \sim 10^1$ - 10^3 keV)

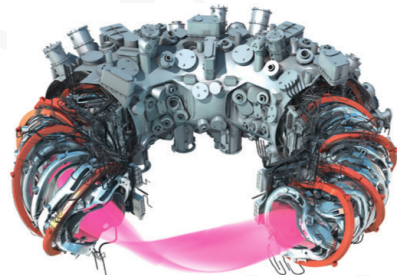
Triple Product

Key figure of merit for fusion performance combining density, temperature, and energy confinement time

WENDELSTEIN 7-X

Key Challenge

Managing heat loads on plasma-facing components during long, high performance plasma

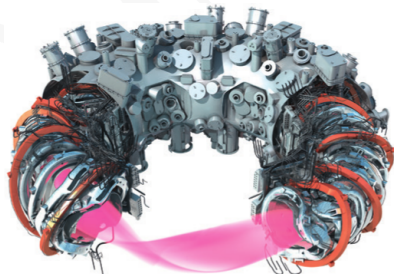


WENDELSTEIN 7-X

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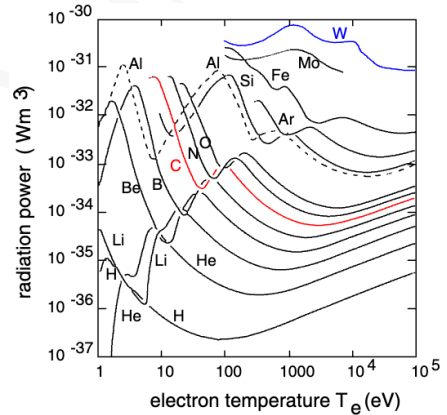
Managing heat loads on plasma-facing components during long, high performance plasma

⇒ Dissipation of large fraction of heat load (energy) in scrape-off layer and particle flux to the target through detachment



PLASMA RADIATION

- plasma profiles so radiation in SOL increases and transport across separatrix reduced

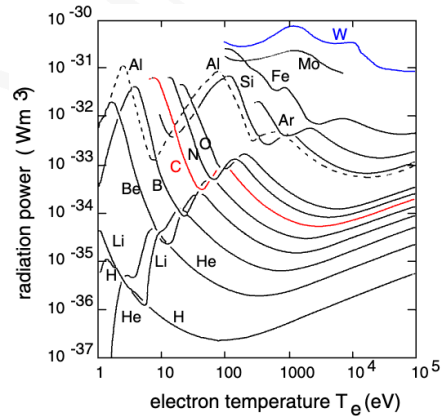


PLASMA RADIATION

- plasma profiles so radiation in SOL increases and transport across separatrix reduced

Impurity Seeding

Injection of low-Z (He, N₂) or high-Z (Ne, Ar) impurities to enhance edge radiation and cooling



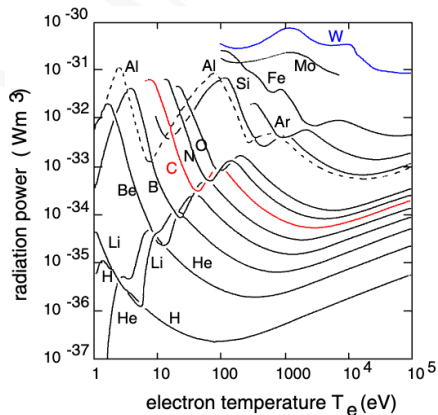
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- plasma profiles so radiation in SOL increases and transport across separatrix reduced
- avoid impurity accumulation

Impurity Seeding

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⇒ stable control w. radiation information?

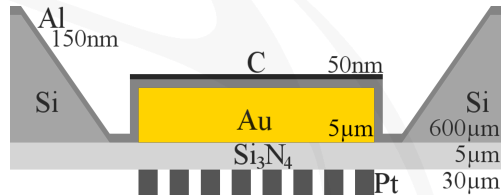


BOLOMETER DIAGNOSTIC

BOLOMETRY AT W7-X

Metal Resistor Bolometers:

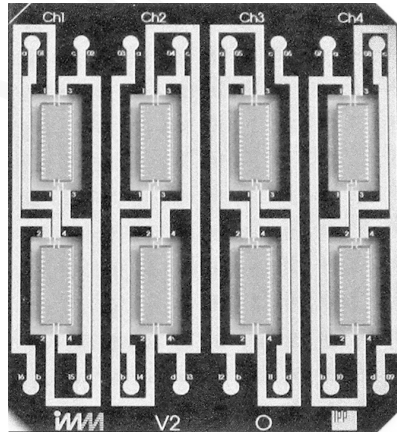
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- Wheatstone bridge circuit configuration
- Measures total radiation power (broadband)
- Multicamera system: HBC & VBCI/r (32 + 2×24 channels)



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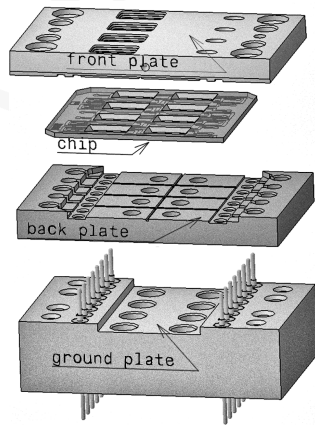
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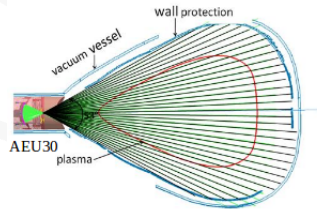
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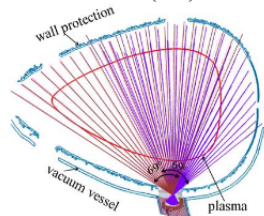
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horizontal bolometer camera (HBC)

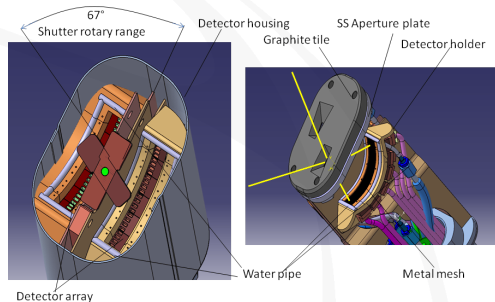


vertical bolometer camera (VBC)

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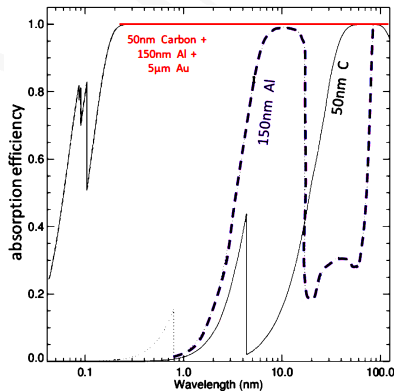
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PERFORMANCE

Performance Characteristics:

- In-situ ohmic heating calibration (F_M , f_M)
- Temporal resolution: $\{0.4, 0.8, 1.6, \dots, 6.4\}$ ms
- Measurement uncertainty (*Gauss*): $1.6 \mu\text{W} \leftrightarrow \text{SNR} > 1000$
- Spatial resolution: ~ 5 cm at magnetic axis
- Near unity absorption efficiency: 600-0.2 nm



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Power Per Detector:

$$P_M = F_M \cdot \left(\frac{d(\Delta \tilde{U}_M)}{dt} + f_M \Delta \tilde{U}_M \right)$$

Global Radiation Power:

$$P_{\text{rad}} = \frac{V_{\text{P,tor}}}{V_C} \sum_M \frac{P_M V_M}{K_M}$$

PERFORMANCE

Chord Brightness

Line-integrated measurement along each detector's line of sight provides radial profile information ($P_{\text{chord}} \sim P_M/K_M$)

Applications:

- Power balance calculations
- **Tomographic reconstruction**
- **Real-time feedback control**

⇒ **controlled radiative cooling in the plasma edge**

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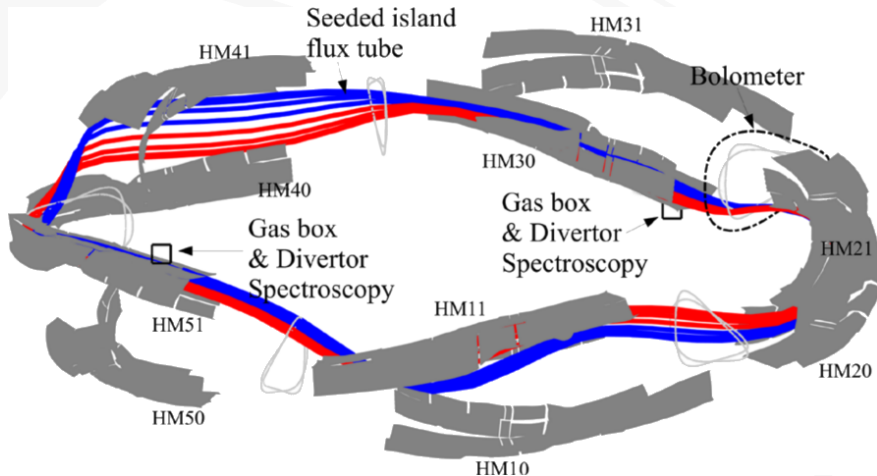
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REAL-TIME RADIATION FEED- BACK CONTROL

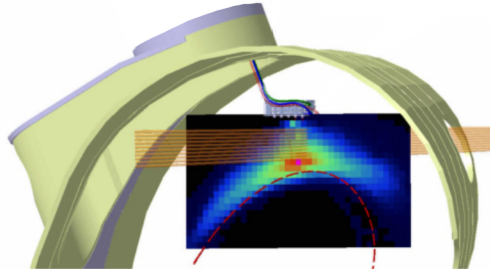
FEEDBACK SYSTEM



FEEDBACK SYSTEM

Architecture:

- Integrated data acquisition hardware & optimized control software
- Twin proxies and dedicated signal outputs
- Thermal gas valve actuation with multipurpose/-diagnostic PID controller



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Radiation Prediction Proxies:

A: *Channel selection S*

$$P_{\text{pred}}^{(1)} = \frac{V_{\text{P,tor}}}{V_S} \sum_M^S \frac{P_M V_M}{K_M}$$

(e.g. $S_{\text{VBC}} = \{61, 86, 78, 73, 70, 65\}$)

B: *Single channel*

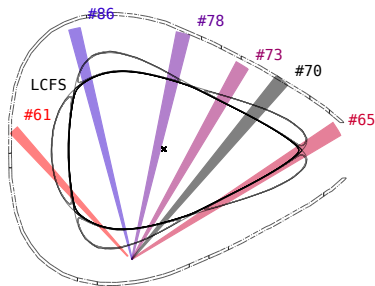
$$P_{\text{pred}}^{(2)} \propto \Delta \widetilde{U}_M$$

(dimensionless, minimal overhead)

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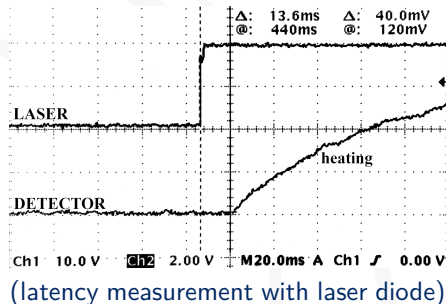
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FEEDBACK SYSTEM

System Performance:

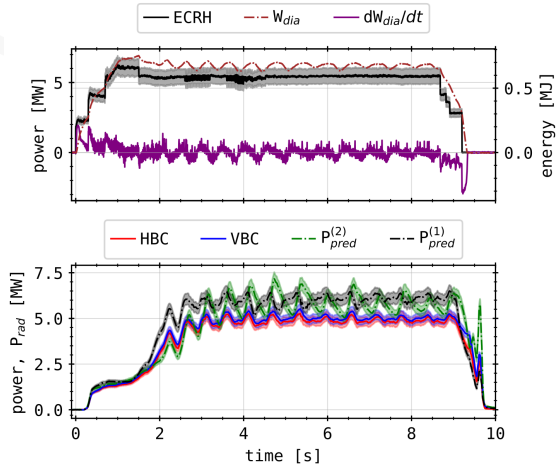
- FIFO smoothing in $P_{\text{pred}}^{(1)}$: $\sim M \cdot \Delta t / 2$ ($M=10$, $\Delta t=1.6$ ms)
- Minimum acquisition latency 13.6 ms
- Total system latency: 150-400 ms (dominated by plasma response)



EXPERIMENTAL ACHIEVEMENTS

XP20181010.32

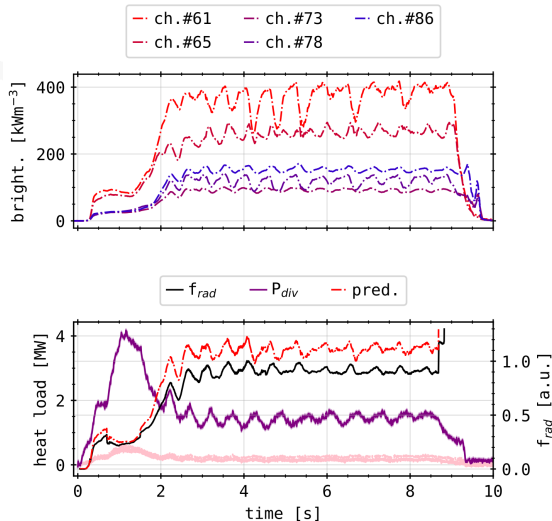
- Duration 9.2 s
 $P_{\text{ECRH}}^{\text{max}} = 6.23 \text{ MW}$ (O2-mode)
- Total input energy 55 MJ
- Radiation fraction PID
 setpoint $f_{\text{rad}} \sim 90\%$ off $P_{\text{pred}}^{(1)}$
- 5-channel VBC subset
 $S = \{65, 78, 61, 73, 86\}$



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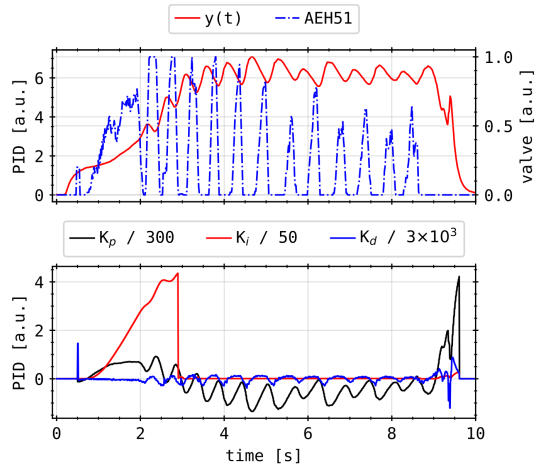
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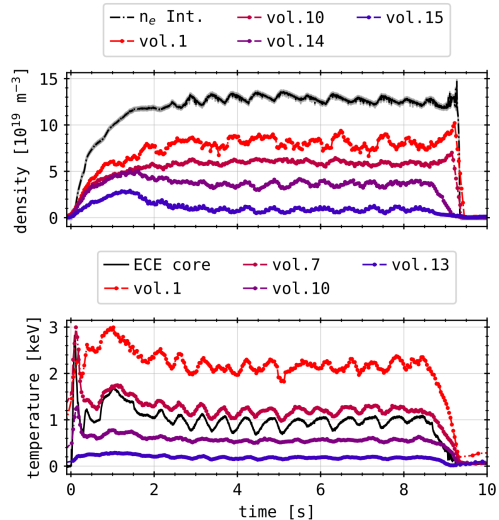
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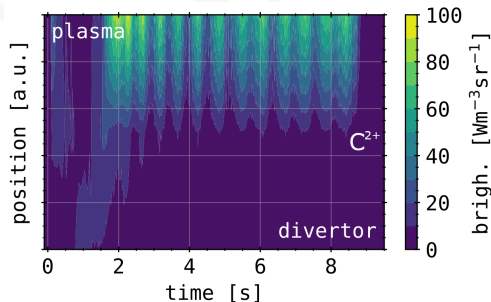
EXPERIMENTAL ACHIEVEMENTS

XP20181010.32

- Peak $f_{\text{rad}} \gtrsim 100\%$, cycling with gas inlet
- Target heat load reduction factor >2
- C^{2+} detachment visible at $f_{\text{rad}} \sim 50\%$

Key Result

Demonstrated stable, real-time feedback-controlled radiative cooling with $f_{\text{rad}} \geq 85\%$ without terminal plasma disruption and significant target heat load reduction



LINE-OF-SIGHT SENSITIVITY

CHANNEL SELECTION SENSITIVITY

- Evaluate LOS/channel efficacy for feedback (generalised local radiation sensitivity)
- Data set of all feedback discharges from OP1.2b
- Applied 5 different metric (linear & weighted deviation, FFT, ...)

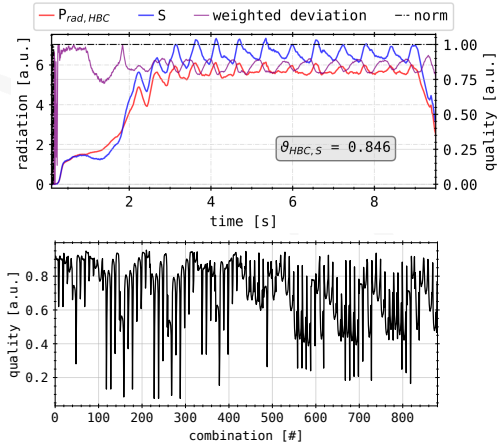
Weighted deviation:

$$\varphi(t) = 1 - \frac{\|P_{\text{pred}} - P_{\text{rad}}\|(t)}{P_{\text{rad}}(t)}$$

$$\vartheta = \frac{1}{T} \int_T \varphi(t) dt$$

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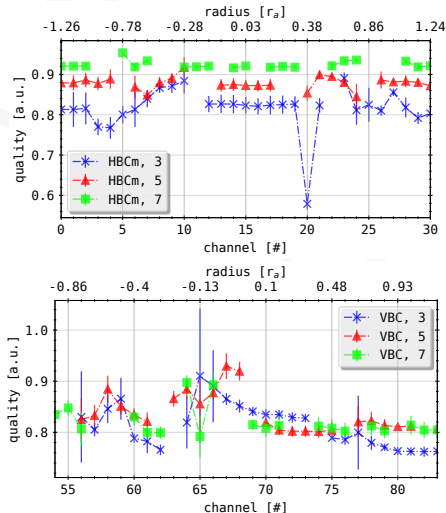
CHANNEL SELECTION SENSITIVITY

Optimal Channel Selection:

- No single "best" set exists
- Robust selection achieves $\geq 85\%$ prediction accuracy
- VBC channels generally outperform HBC for edge sensitivity
- 3-5 channels sufficient for reliable P_{pred}

Takeaway

Detectors viewing close to separatrix and SOL most viable for feedback purposes



TOMOGRAPHY

MINIMUM FISHER REGULARIZATION

Measurement:

$$\mathbf{T} \vec{x} = \vec{b}$$

Bolometer tomography:

- 61 line-integrated measurements $\vec{b}$
- 2D emissivity $\vec{x}$: ~ 4500 pixels
- *Geometry* $\mathbf{T}$ has high condition
- **ill-posed, requires regularization**

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Problem:

$$\min_{\eta} \left\| \left(\mathbf{T}^T \mathbf{T} + \mu^{-1} \mathbf{H} \right)^{-1} \mathbf{T}^T \vec{b} \right\| \rightarrow \vec{x}$$

Fisher Information:

$$I_F = \int \frac{1}{x} \left(\frac{\partial x}{\partial \vec{r}} \right)^2 d\vec{r} \leftrightarrow \mathbf{H} \sim \frac{1}{x} \Delta^T \Delta$$

limiting error on reconstruction:

$$\sigma_x \leq \frac{1}{I_F}$$

MINIMUM FISHER REGULARIZATION

MFR + RDA Advantages

- Robust to noisy data
- Incorporates a priori knowledge
- Stable convergence

Radially Dependent Anisotropy:

- Tailored weighting to poloidal gradient
- $k_{\text{ani}} < 1$: favours anisotropy
- $k_{\text{ani}} > 1$: favours smooth emissivity

$$\mathbf{H} \rightarrow \mathbf{H}_{\text{ani}}$$

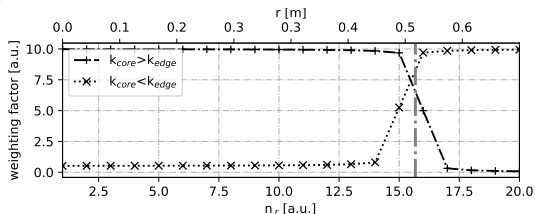
$$\Delta = \nabla_{\mathbf{r}}^{\mathbf{T}}(\cdot)\nabla_{\mathbf{r}} + \nabla_{\vartheta}^{\mathbf{T}}(\cdot)\nabla_{\vartheta}$$

$$\nabla_{\vartheta} \rightarrow \tilde{\nabla}_{\vartheta} = \frac{1}{k_{\text{ani}}\Delta r} \mathbf{D}_{\text{dia}}^{(0)} \nabla_{\vartheta}$$

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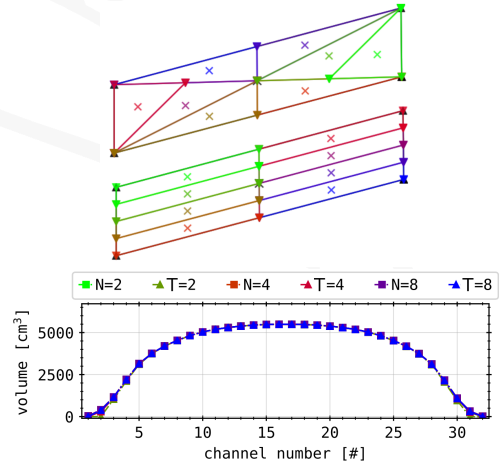
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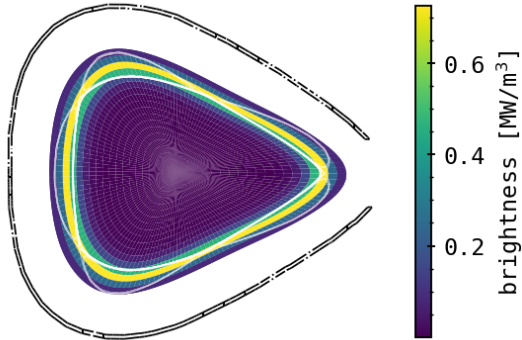
BENCHMARKING

Geometry Perturbation Test:

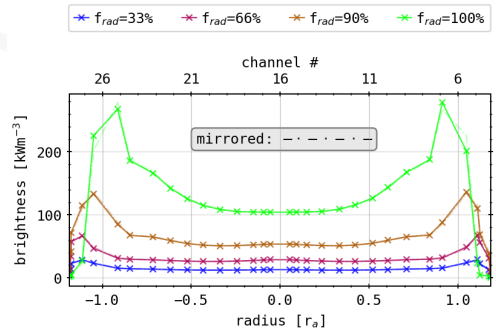
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BENCHMARKING



($f_{\text{rad}} \sim 100\%$)

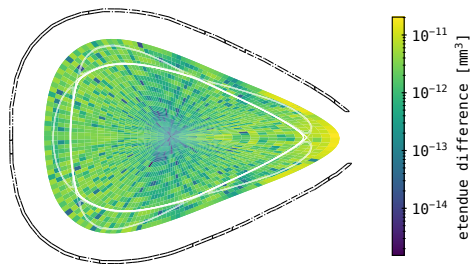
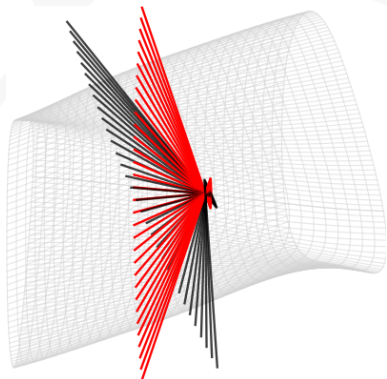


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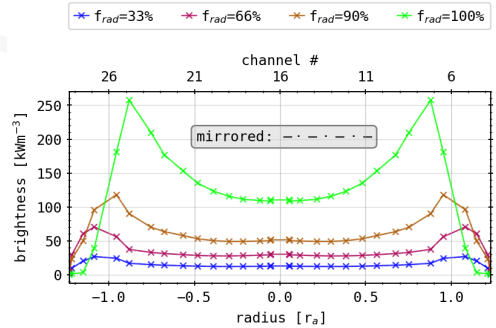
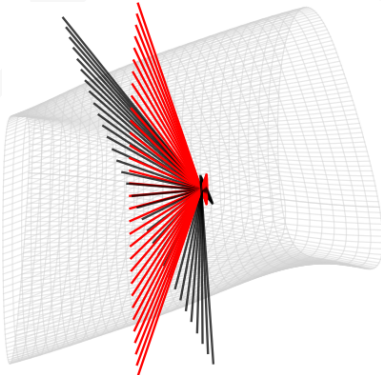
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- Tests for symmetrical chord profile yield artificial camera geometry

BENCHMARKING



$$\|T - T_{\text{sym}}\|$$

BENCHMARKING

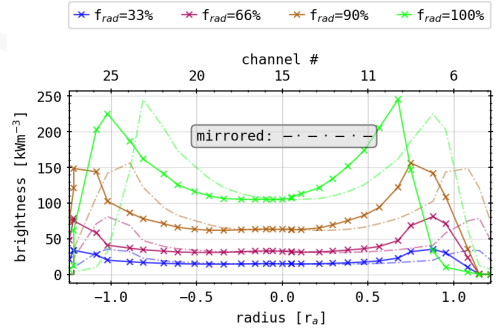
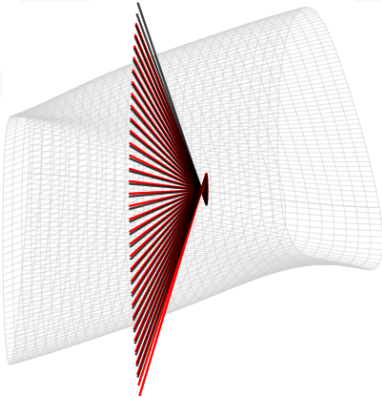


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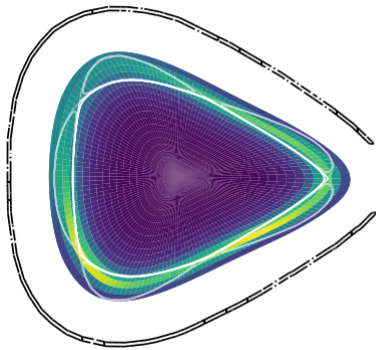
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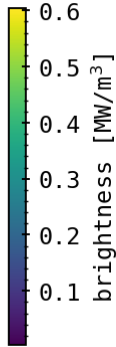
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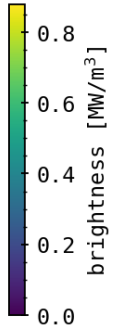
BENCHMARKING



using T_{sym}



using T_{ilt}



BENCHMARKING

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Results

- Discretization of system is computationally stable
- Intrinsic asymmetry of same order of magnitude as potential, unknown geometric perturbation - non-negligible
- Worst case machine warp scenario $\sim 0.2r_a$ shift

PHANTOM RECONSTRUCTIONS

Purpose

Regularized, parametric inversion methods need (experienced) tailoring towards their respective application - if not characteristic distributions.

PHANTOM RECONSTRUCTIONS

Benchmarks:

- 12+ synthetic emissivities: Gaussian, rings, anisotropies, convolutions...
- k_{ani} & phantom variations studies (radius, intensity etc.)
- Experimentally motivated distributions
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Metrics:

$$\chi^2 = \frac{1}{N_{\text{ch}}} \left\| \left(\vec{b}_{\text{phan}} - \vec{b}_{\text{tom}} \right) \vec{\sigma}_{\text{bol}}^{-1} \right\|$$

$$\rho_c = \frac{1}{\sigma_{\text{pha}} \sigma_{\text{tom}}} \text{cov}(\vec{x}_{\text{phan}}, \vec{x}_{\text{tom}})$$

$$M_{\text{SD}} \sim \left\| \frac{\vec{x}_{\text{tom}} - \vec{x}_{\text{phan}}}{\|\vec{x}_{\text{phan}}\|} \cdot \vec{A}^{\text{T}} \right\|$$

$$P_{\text{rad},X}^{2\text{D}} = \vec{x}_X^{\text{T}} \cdot \vec{A}, \quad X = \{\text{phan}, \text{tom}\}$$

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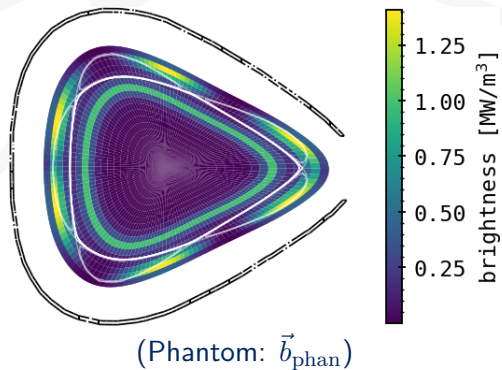
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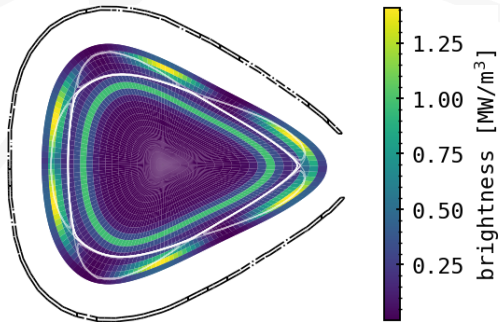
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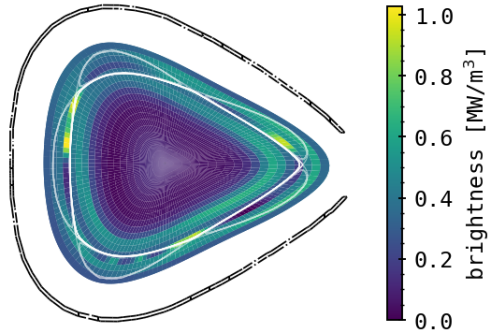
PHANTOMS: EXAMPLE



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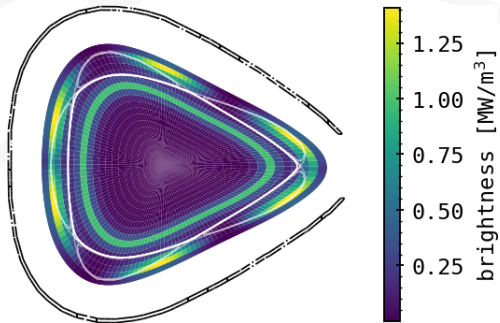


(Phantom: $\vec{b}_{\text{phan}}$)

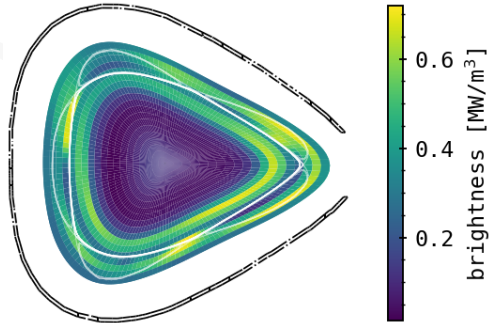


$k_{\text{core}}, k_{\text{edge}} = \{2, 0.1\}$

PHANTOMS: EXAMPLE

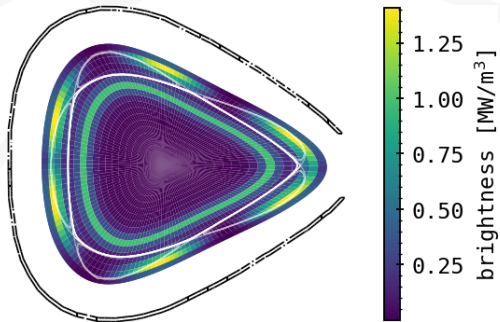


(Phantom: $\vec{b}_{\text{phan}}$)

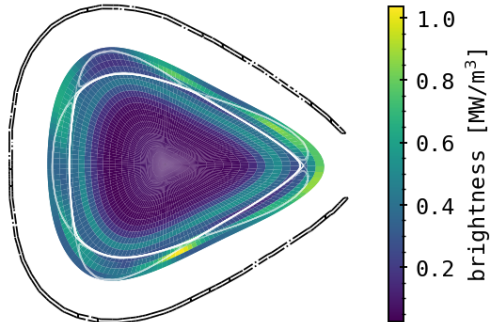


$k_{\text{core}}, k_{\text{edge}} = \{2, 0.3\}$

PHANTOMS: EXAMPLE

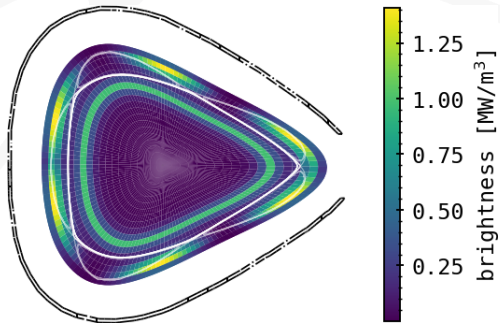


(Phantom: $\vec{b}_{\text{phan}}$)

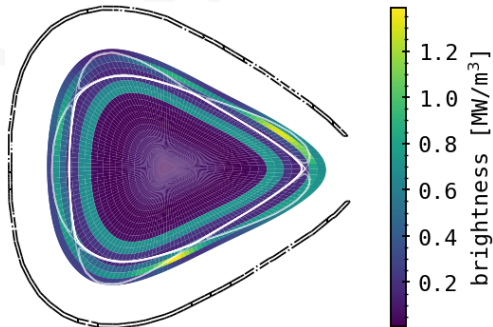


$k_{\text{core}}, k_{\text{edge}} = \{2, 0.6\}$

PHANTOMS: EXAMPLE

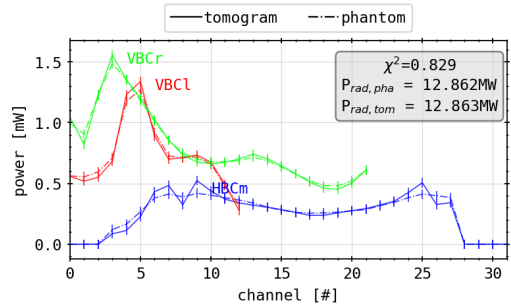
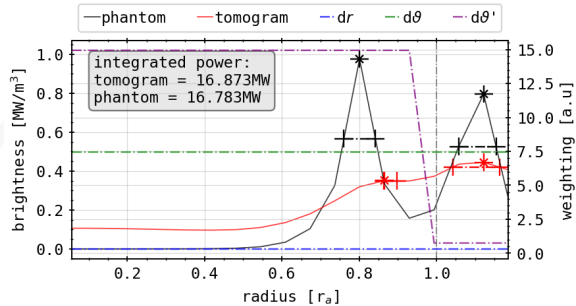


(Phantom: $\vec{b}_{\text{phan}}$)

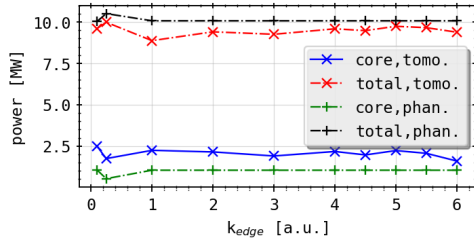
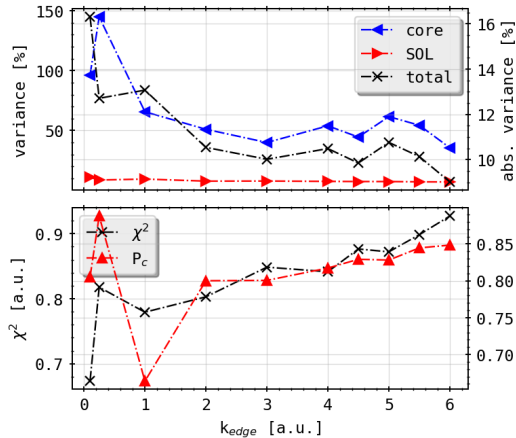


$k_{\text{core}}, k_{\text{edge}} = \{20, 3.0\}$

PHANTOMS: EXAMPLE

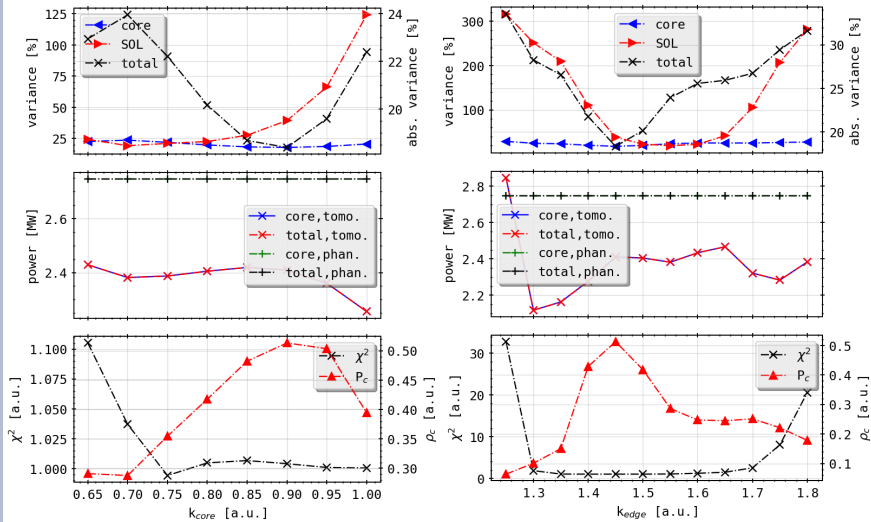


PHANTOMS: EXAMPLE

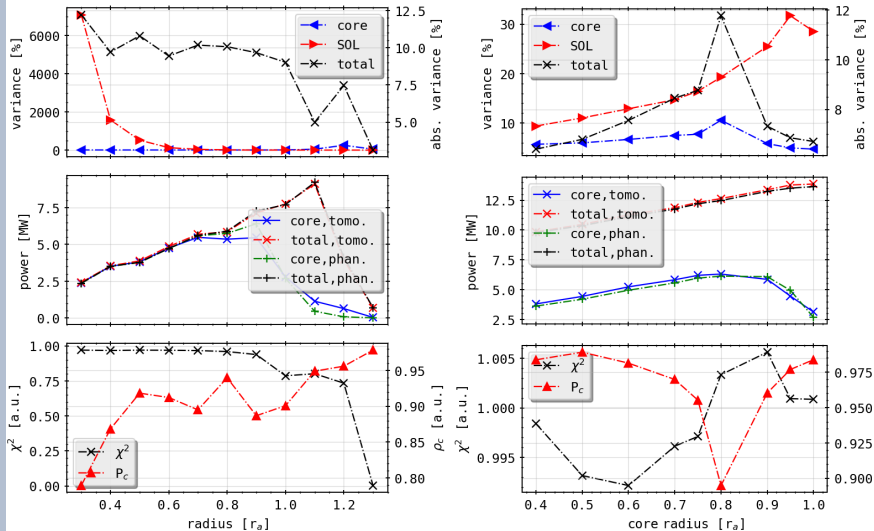


PHANTOMS: STATISTICS

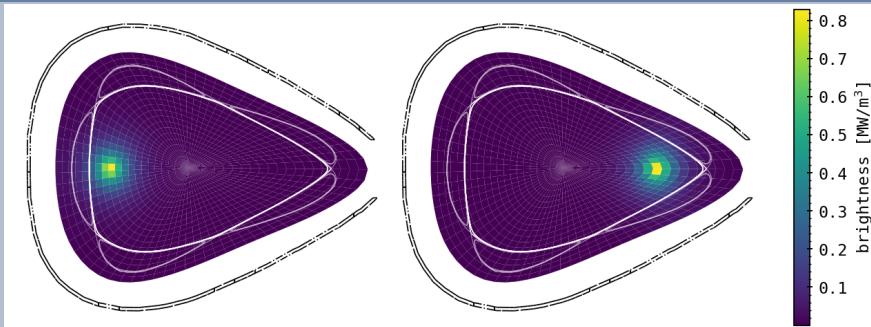
Isotropic Rings: k_{ani} scan



Isotropic Rings: Radial Scan



Composite Distributions



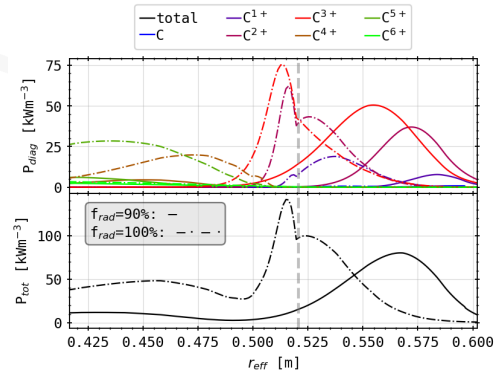
Composite Distributions



STRAHL IMPURITY TRANSPORT MODELING

STRAHL Code:

- 1D impurity transport and radiation simulation
- Solves radial continuity for each ionisation stage
- Experimental input data: perturbation study in transport coefficients, profile smoothing, source model etc.
- Valid in confinement, simple ansatz outside LCFS



STRAHL IMPURITY TRANSPORT MODELING

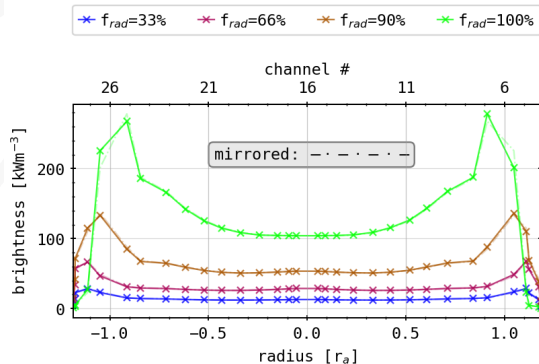
STRAHL Code:

- 1D impurity transport and radiation simulation
- Solves radial continuity for each ionisation stage
- Experimental input data: perturbation study in transport coefficients, profile smoothing, source model etc.
- Valid in confinement, simple ansatz outside LCFS

Key Results:

- Carbon dominates emissivity ($C^{2/3+}$, $\times 10^2+$ of Oxygen levels)
- Inward shift of radiation for $f_{\text{rad}} \rightarrow 1$ toward $r \sim 0.95r_a$
- Radiation moves from outside to inside separatrix

STRAHL IMPURITY TRANSPORT MODELING



Forward Modeling:

- Chord brightness from STRAHL results
- Compare with experimental measurements
- Validate assumptions about impurity radiation and transport

Discrepancy

Forward calculations show asymmetry - less than experiments - in confinement, suggesting intrinsic geometry effects

STRAHL IMPURITY TRANSPORT MODELING

